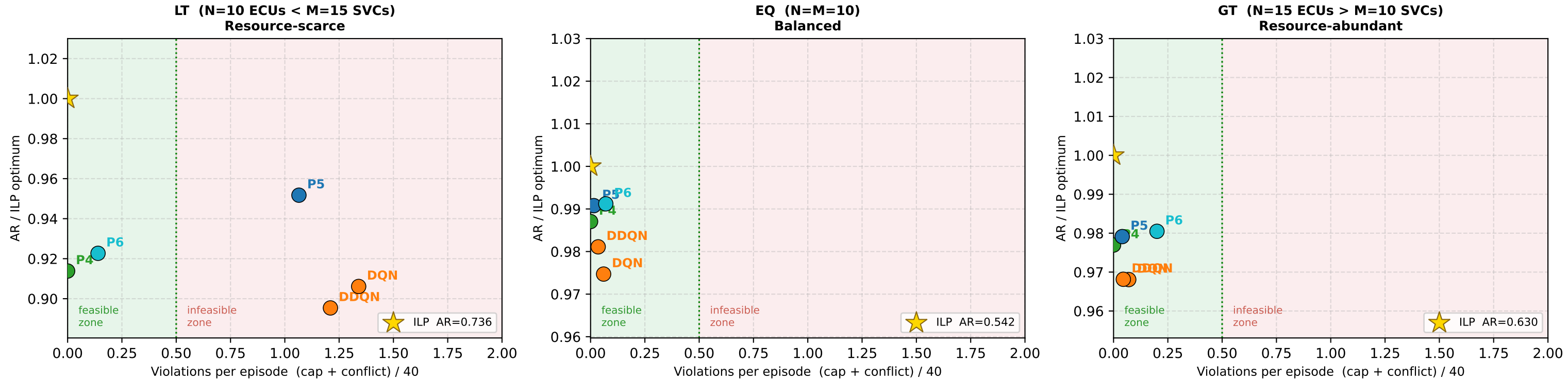


Safety-Performance Frontier: AR (% of ILP) vs Violations per Episode
Green zone = practically feasible (< 0.5 violations/episode); red zone = unacceptable constraint violations



Legend:

- P4 — Hard mask (zero violations guaranteed)
- P5 — Lagrangian / soft constraint
- P6 — Repair heuristic
- DQN
- DDQN